E-commerce Sales Analysis

Pivot Tables and Dashboard Project

Dataset Overview

The workbook has four sheets that together cover two years of orders (October 2023 to October 2025).

Sheet	Rows	Content
Sales_Fact	2,000	Sales_ID, Order_Date, Customer_ID, Product_ID, Store_ID, Quantity, Unit_Price, Discount, Payment_Type, Total_Amount
Customer_Dim	500	Age, Gender, City, State, Country
Product_Dim	100	Product name, Category, Sub-Category, Cost, Stock
Store_Dim	20	Region, City, Store_Type (Flagship, Outlet, Online)

Problem Statement

The company has two years of transaction data but no clear view of what drives its revenue and profit. It does not know which products, stores and regions perform best, how discounts affect sales and profit, or which customer groups buy the most. This project cleans the data, builds pivot tables and charts, and presents the findings in an interactive Excel dashboard that supports decisions on pricing, stock and store strategy.

Objectives of the Excel Analysis

The goal of this project is to:

- Clean and prepare data from four related sheets
- Summarize large datasets using Pivot Tables
- Visualize trends using Pivot Charts
- Build relationships between sheets and calculate Profit using lookup functions
- Create a dynamic and interactive dashboard with slicers
- Present business insights and recommendations

Business Questions to Answer

1. Overall Sales Performance

- What is the total revenue, total profit and total units sold?
- What is the monthly and quarterly sales trend over the two years?
- Which month recorded the highest and lowest revenue?

2. Product Performance Analysis

- Which categories and sub-categories contribute the highest revenue and profit?
- Which are the top 10 and bottom 10 products by revenue?
- Which products have high sales volume but low revenue or profit?

3. Store and Store-Type Performance

- Which store generates the highest revenue?
- How do Flagship, Outlet and Online stores compare in revenue and profit?
- Are there stores with strong unit sales but lower revenue efficiency?

4. Regional Analysis

- How does revenue vary across the North, South, East and West regions?
- Which region contributes the maximum share of total revenue?
- Identify region-wise category preferences.

5. Discount and Pricing Analysis

- What is the average selling price per category?
- Do higher discounts increase the quantity sold, or only reduce revenue and profit?
- Which products or orders are sold at a loss after discount?

6. Customer and Payment Analysis

- Which age groups and gender segments generate the highest revenue?
- Which states have the most valuable customers?
- Which payment type (Credit Card, PayPal, COD) is used most, and which has the highest average order value?

7. Inventory Analysis

- Which products have high stock but low sales (overstocked)?
- Which products have low stock but high sales (risk of stock-out)?

Excel Tools and Techniques to Be Used

1. Data Cleaning

- Find & Replace to standardize IDs (for example, P- to PROD, C- to CUST)
- IF, ROUND and MEDIAN functions to fill missing values
- PROPER and TRIM to fix customer names
- Remove Duplicates and filters to check for blanks

2. Data Modelling

- Excel Tables for structured references
- XLOOKUP or VLOOKUP to bring Category, Region, Store_Type and Cost into Sales_Fact
- Calculated column for Profit = Total_Amount - (Quantity x Cost)

3. Pivot Tables

- Summaries by month, category, sub-category, region, store type, payment type and age group
- Grouping of dates (months, quarters) and of discount ranges
- Top 10 value filters for products

4. Pivot Charts

- Line chart for the monthly trend
- Column and bar charts for category, store and region comparison
- Pie or stacked chart for payment type and revenue share

5. Dashboard

- Slicers for Region, Category and Store_Type, plus a Timeline for date
- KPI cells for Total Revenue, Total Profit, Total Orders and Average Order Value
- A final layout combining all charts on one sheet

Expected Outcome

A cleaned dataset, a set of pivot tables and charts, and one interactive dashboard that shows the key drivers of revenue and profit. The report ends with business insights and recommendations on discounting, stock management and store performance.

Screenshots and Analysis Steps

Step 1: Data Cleaning

In this step I cleaned the four sheets before using them. I used Find and Replace to fix the IDs and the country name, and PROPER, TRIM and CLEAN to fix the names and cities. The missing values in Quantity, Unit_Price, Discount, Total_Amount, Cost and Stock were filled with IF and MEDIAN formulas.

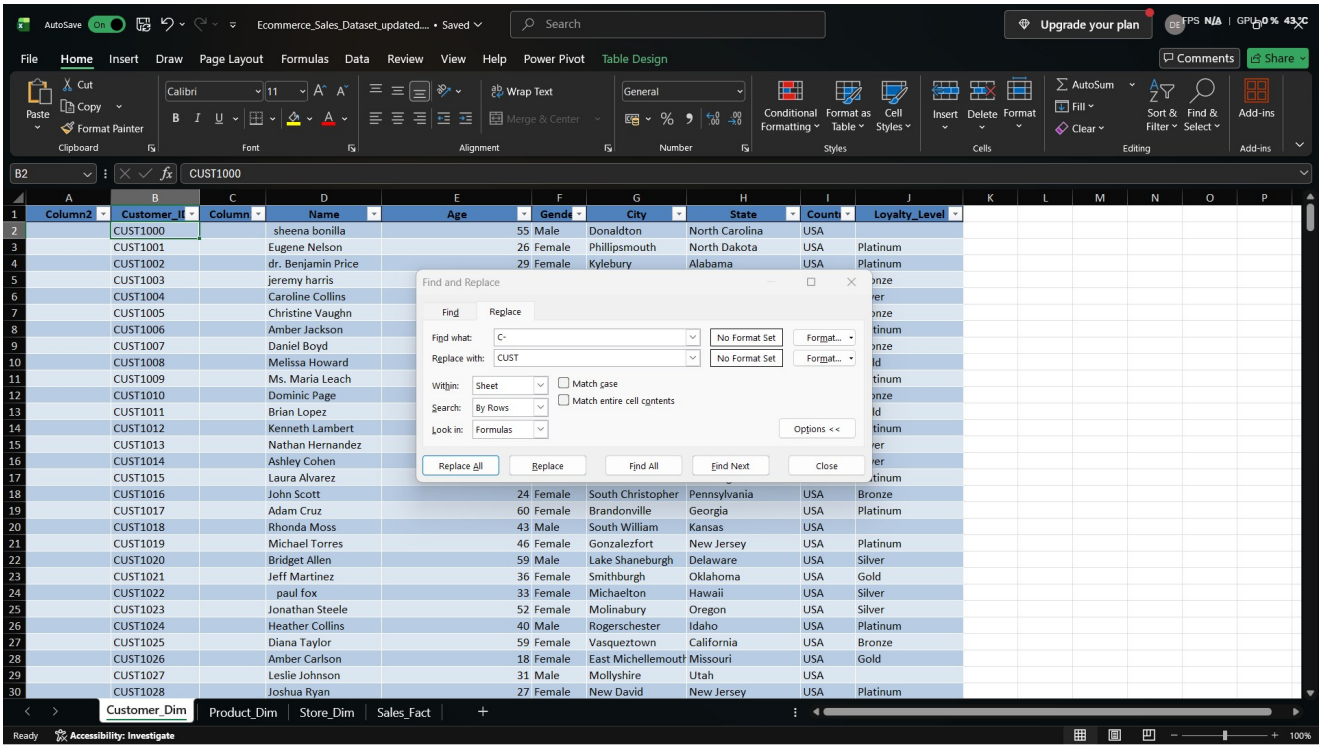


Figure 1: Find and Replace used to change the Customer_ID format (C- to CUST)

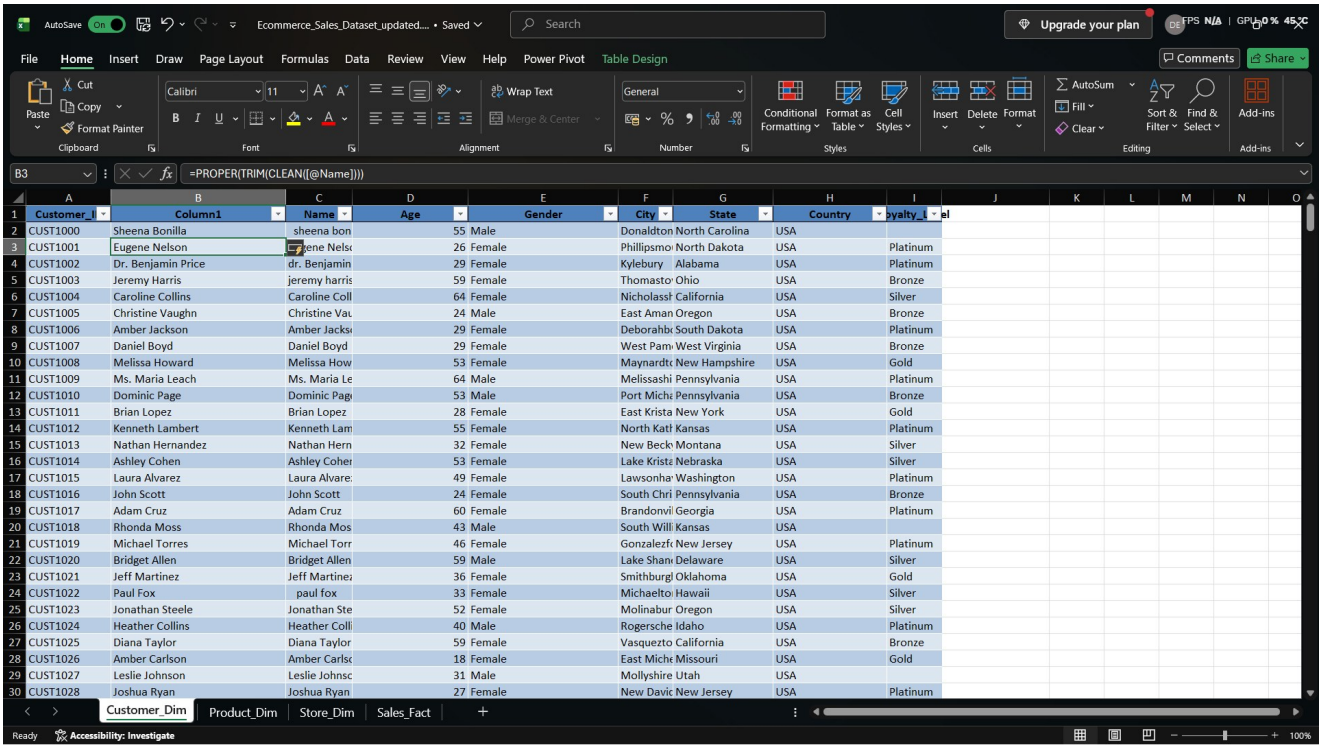


Figure 2: PROPER, TRIM and CLEAN used to fix the customer names

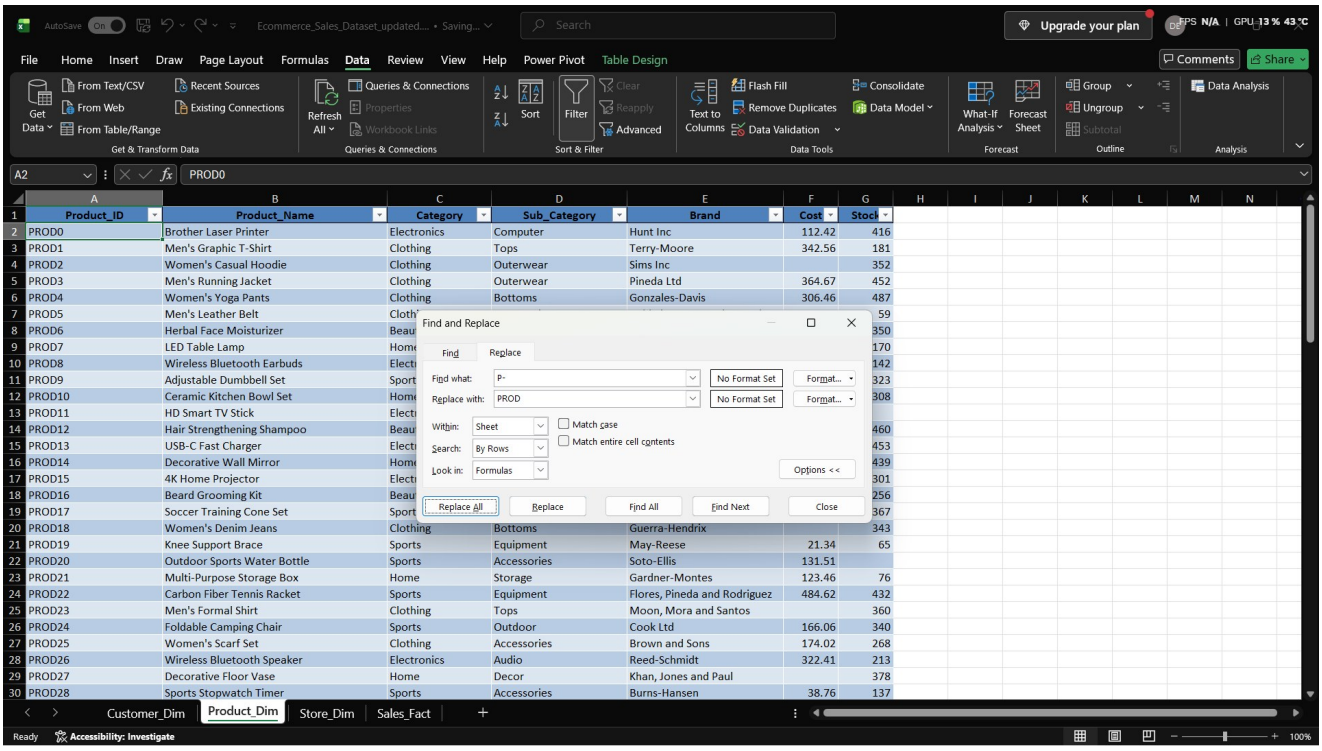


Figure 3: Find and Replace used to change the Product_ID format (P- to PROD)

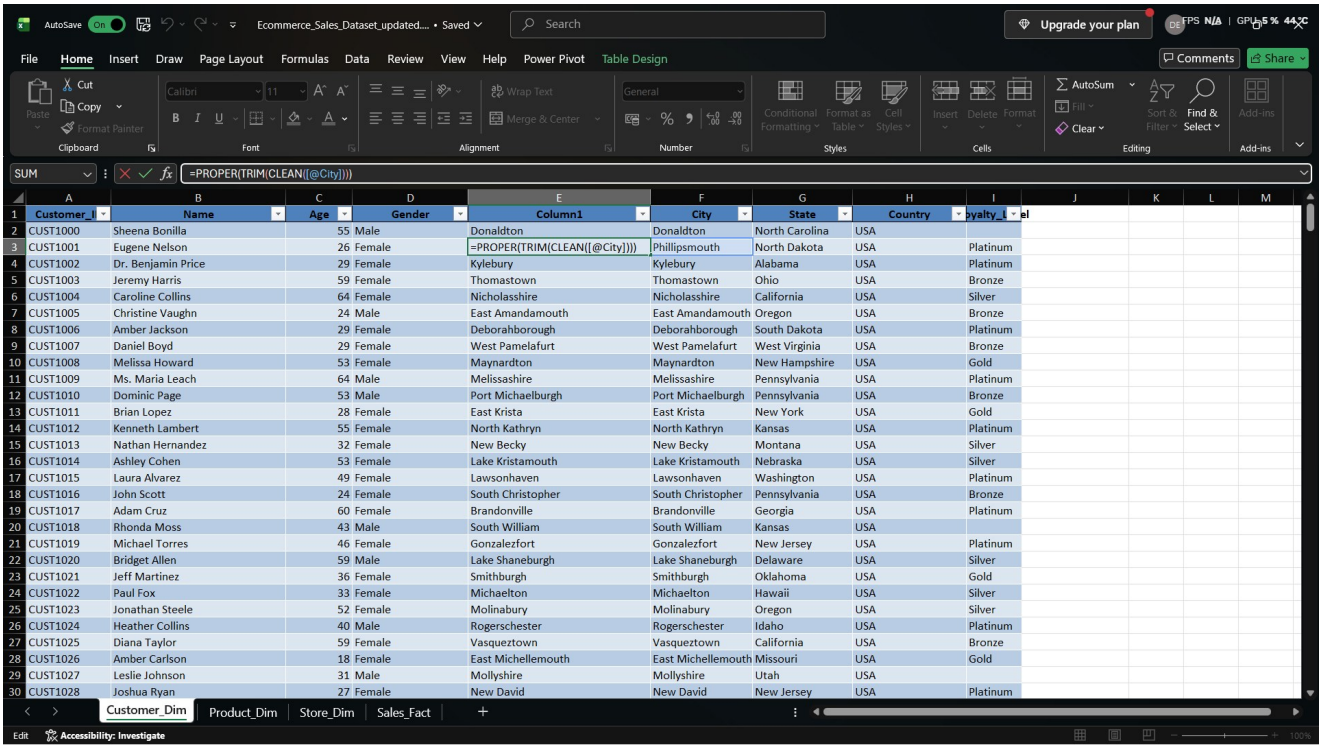


Figure 4: PROPER, TRIM and CLEAN used to fix the city names

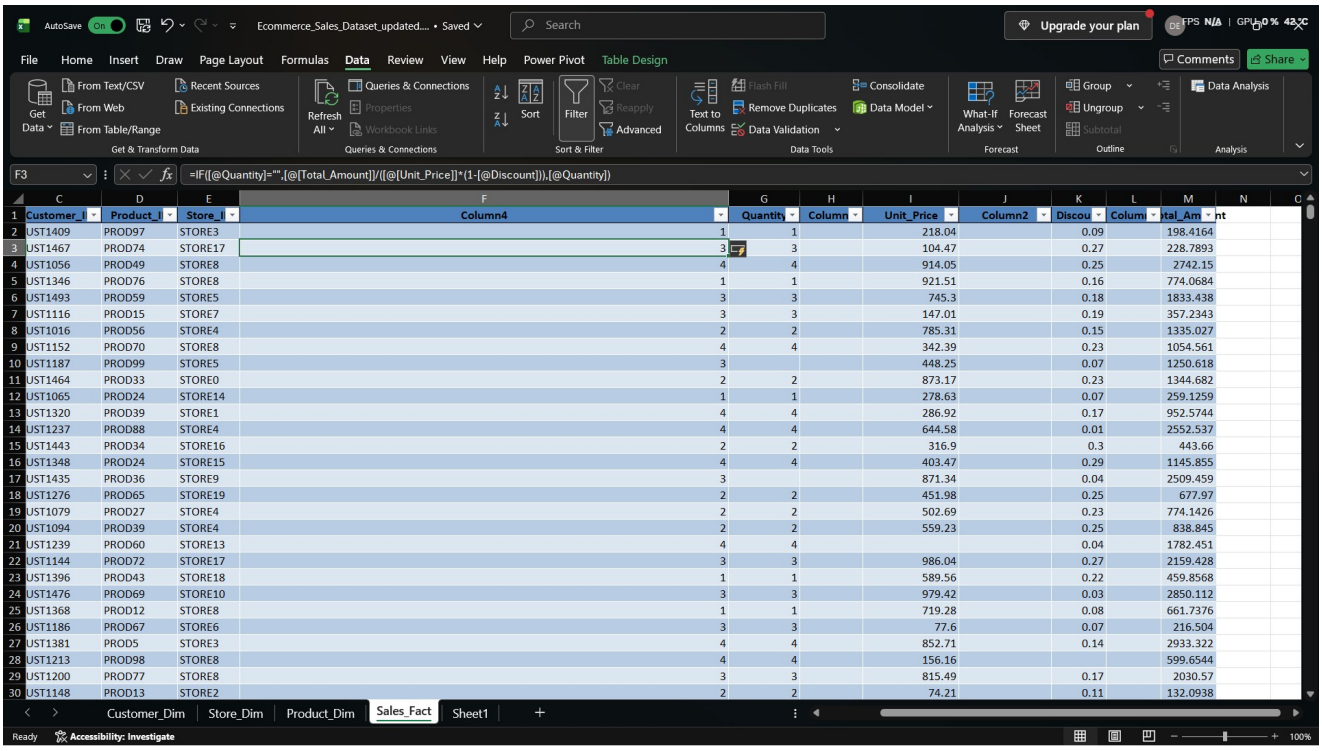


Figure 5: Formula to fill the missing Quantity values in Sales_Fact

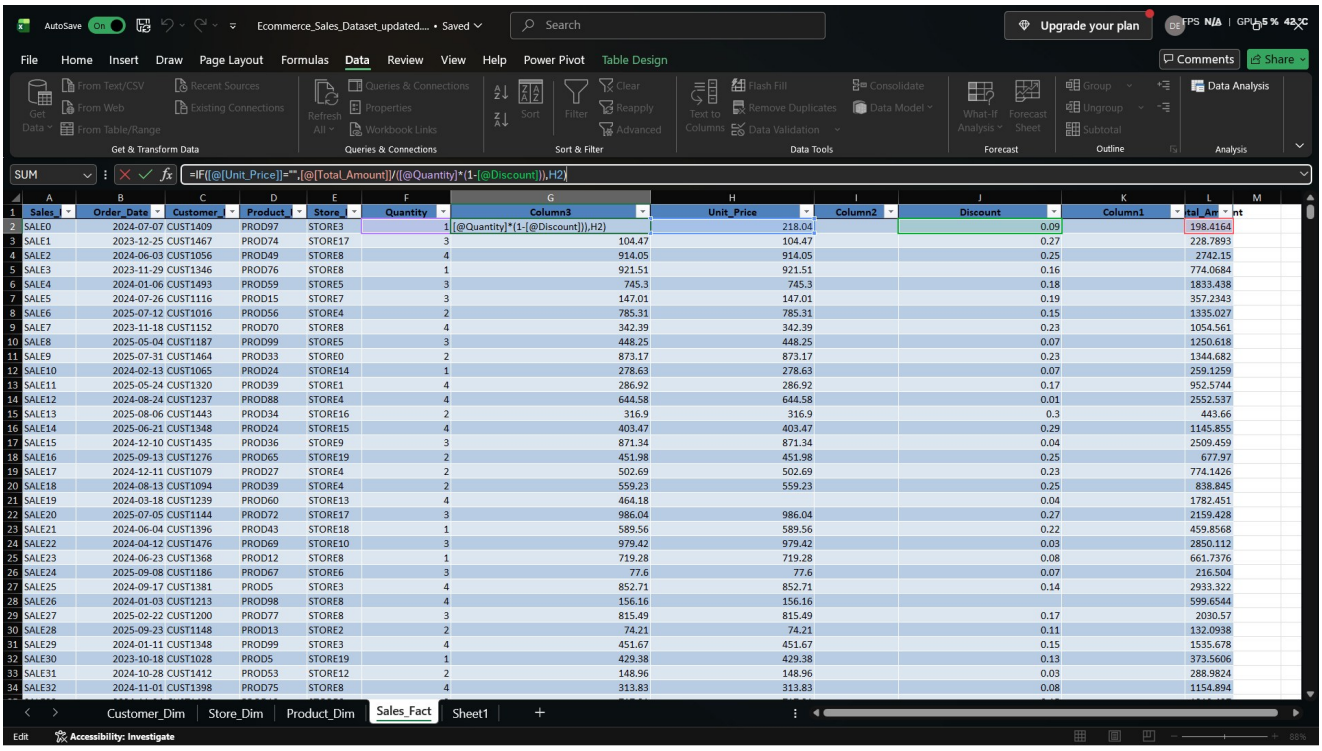


Figure 6: Formula to fill the missing Unit_Price values in Sales_Fact

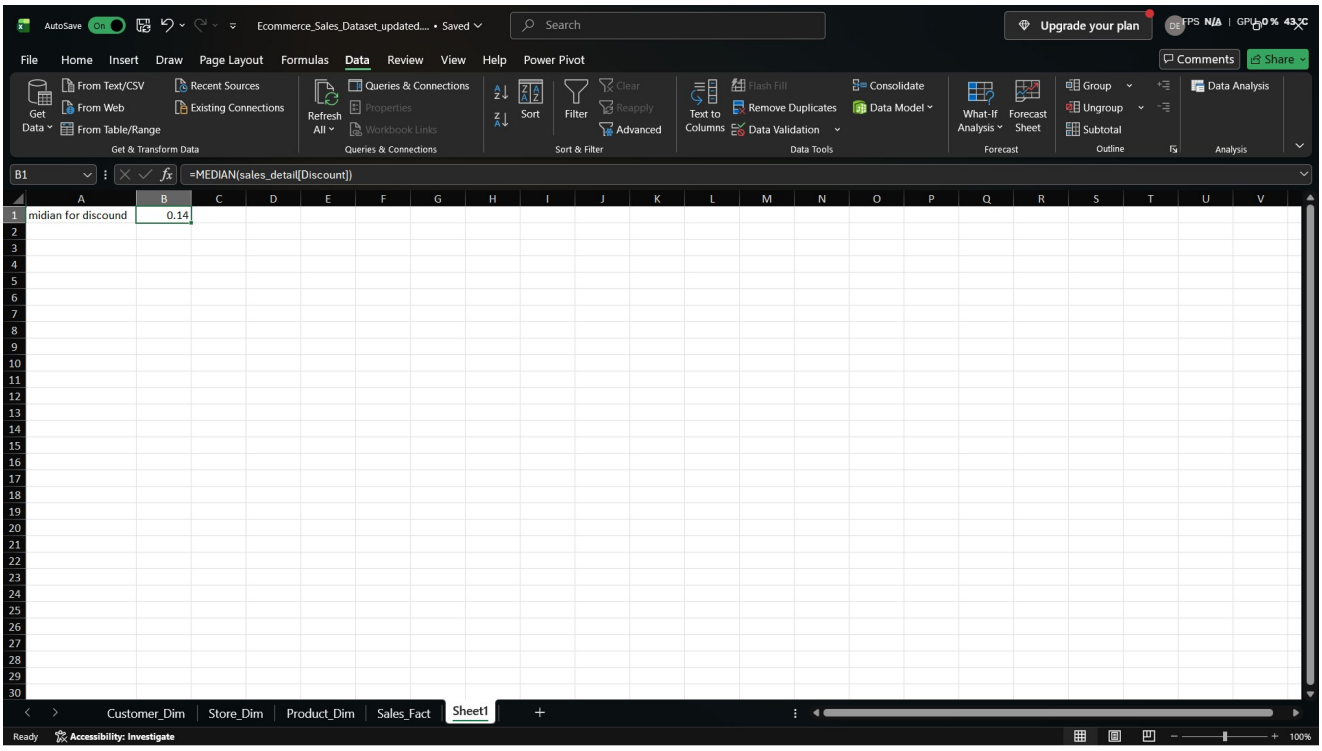


Figure 7: Median of the Discount column calculated in Sheet1 (0.14)

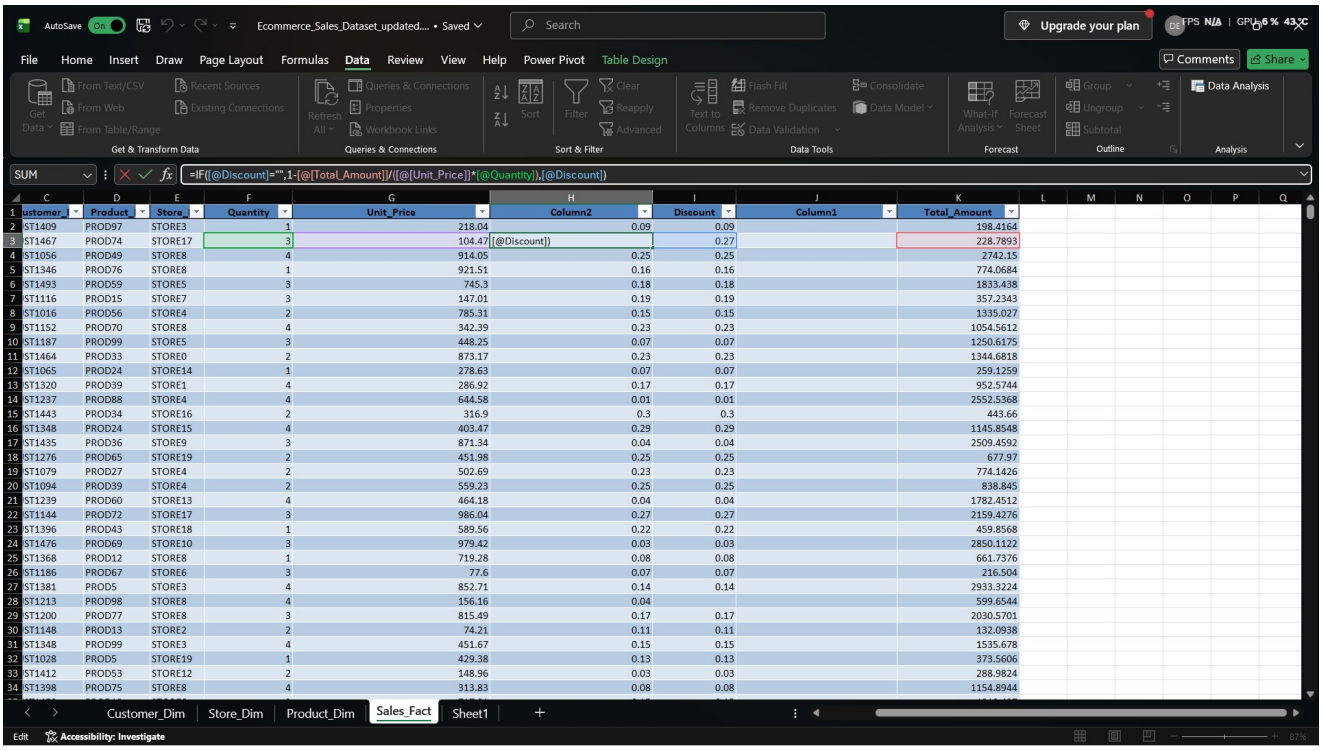


Figure 8: Missing Discount values filled using the median

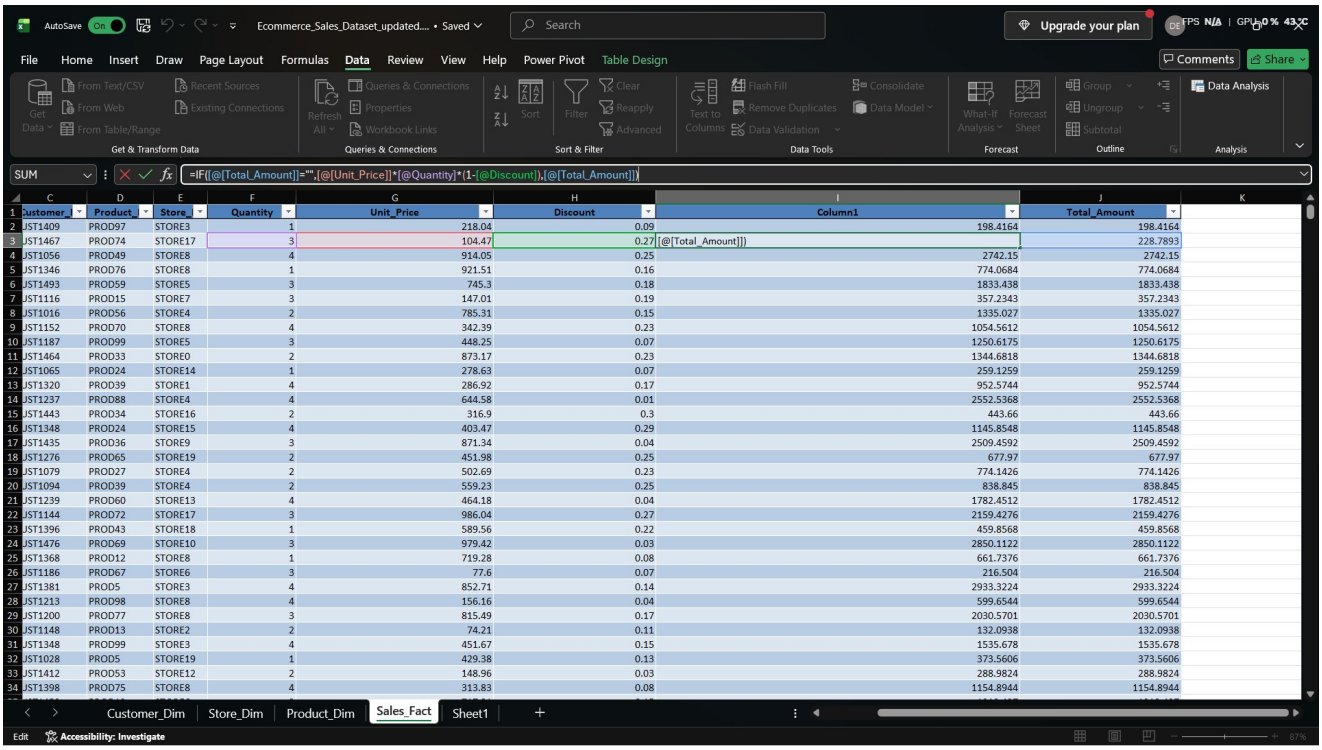


Figure 9: Formula to fill the missing Total_Amount values

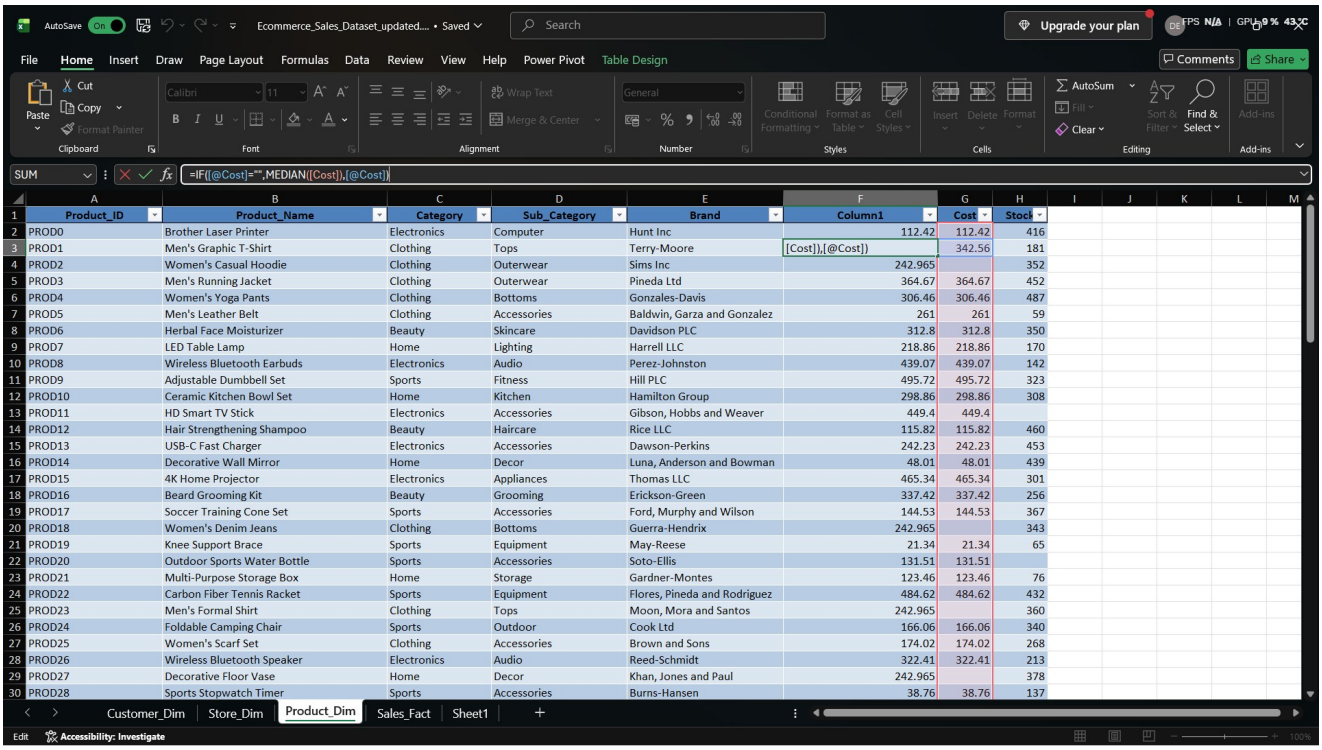
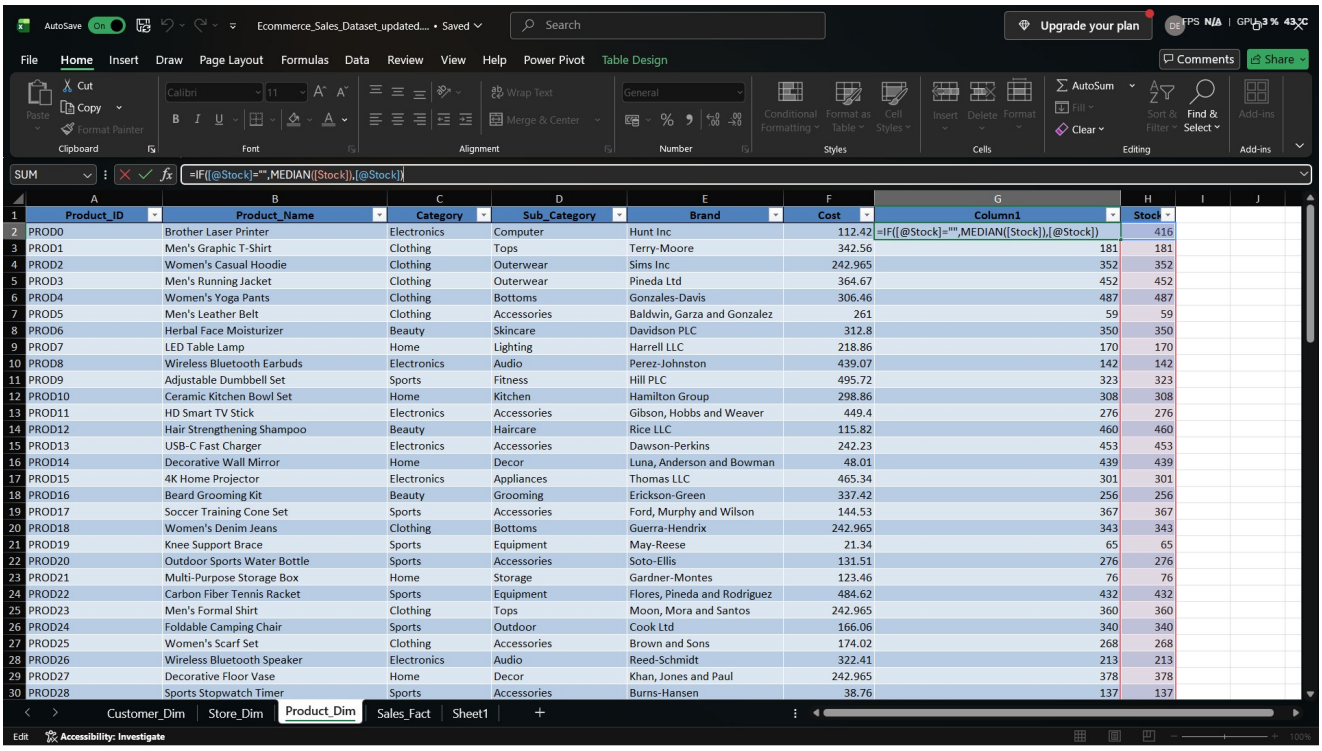
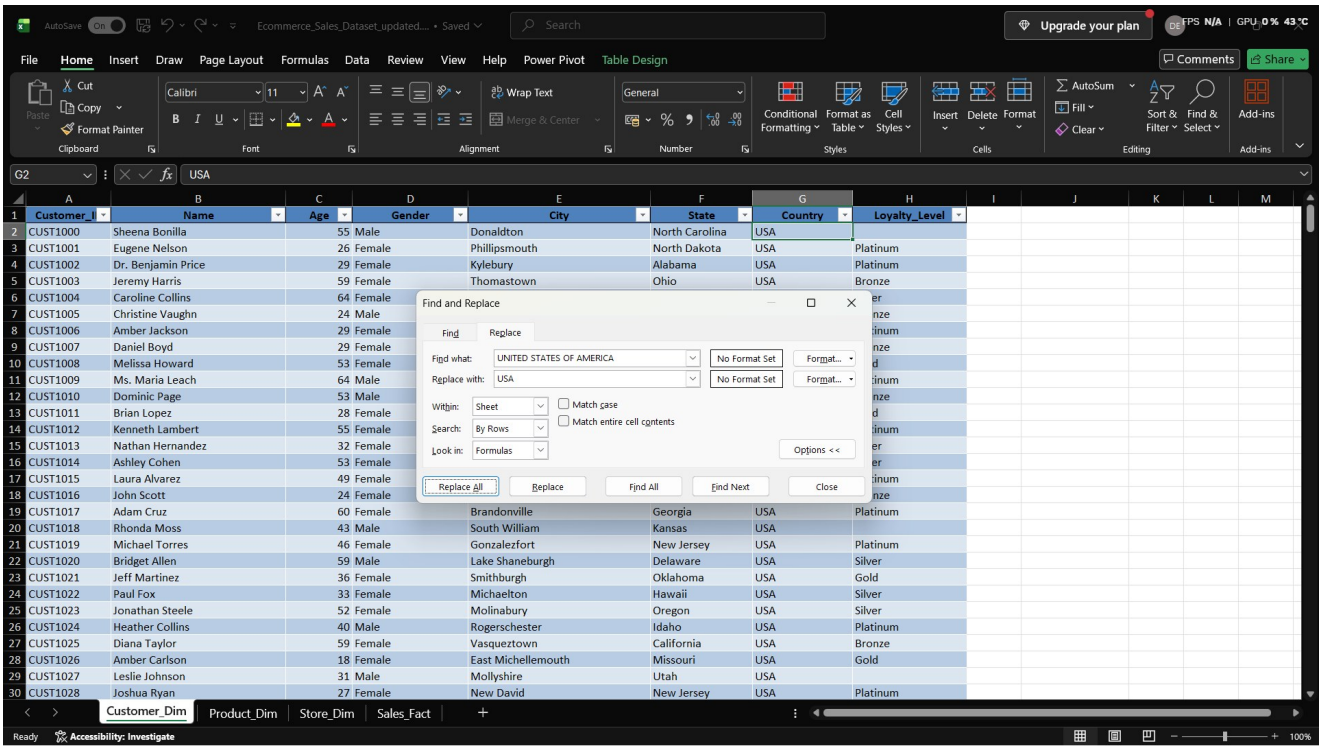


Figure 10: Missing Cost values in Product_Dim filled using the median



Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Column1	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	=IF([@Stock]="",MEDIAN([Stock]),[@Stock])	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	242.965	352	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	276	276
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	242.965	343	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	276	276
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	242.965	360	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	322.41	213	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	242.965	378	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	38.76	137	137

Figure 11: Missing Stock values in Product_Dim filled using the median



Customer_I	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female				
CUST1005	Christine Vaughn	24	Male				
CUST1006	Amber Jackson	29	Female				
CUST1007	Daniel Boyd	29	Female				
CUST1008	Melissa Howard	53	Female				
CUST1009	Ms. Maria Leach	64	Male				
CUST1010	Dominic Page	53	Male				
CUST1011	Brian Lopez	28	Female				
CUST1012	Kenneth Lambert	55	Female				
CUST1013	Nathan Hernandez	32	Female				
CUST1014	Ashley Cohen	53	Female				
CUST1015	Laura Alvarez	49	Female				
CUST1016	John Scott	24	Female				
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum

Figure 12: Country name changed from United States of America to USA

Step 2: Data Modelling

Here I connected the four sheets using the ID columns. Sales_Fact is linked to Customer, Product and Store in the Data Model. I also used XLOOKUP to bring product name and customer name into the sales sheet, and added a total row for Total_Amount.

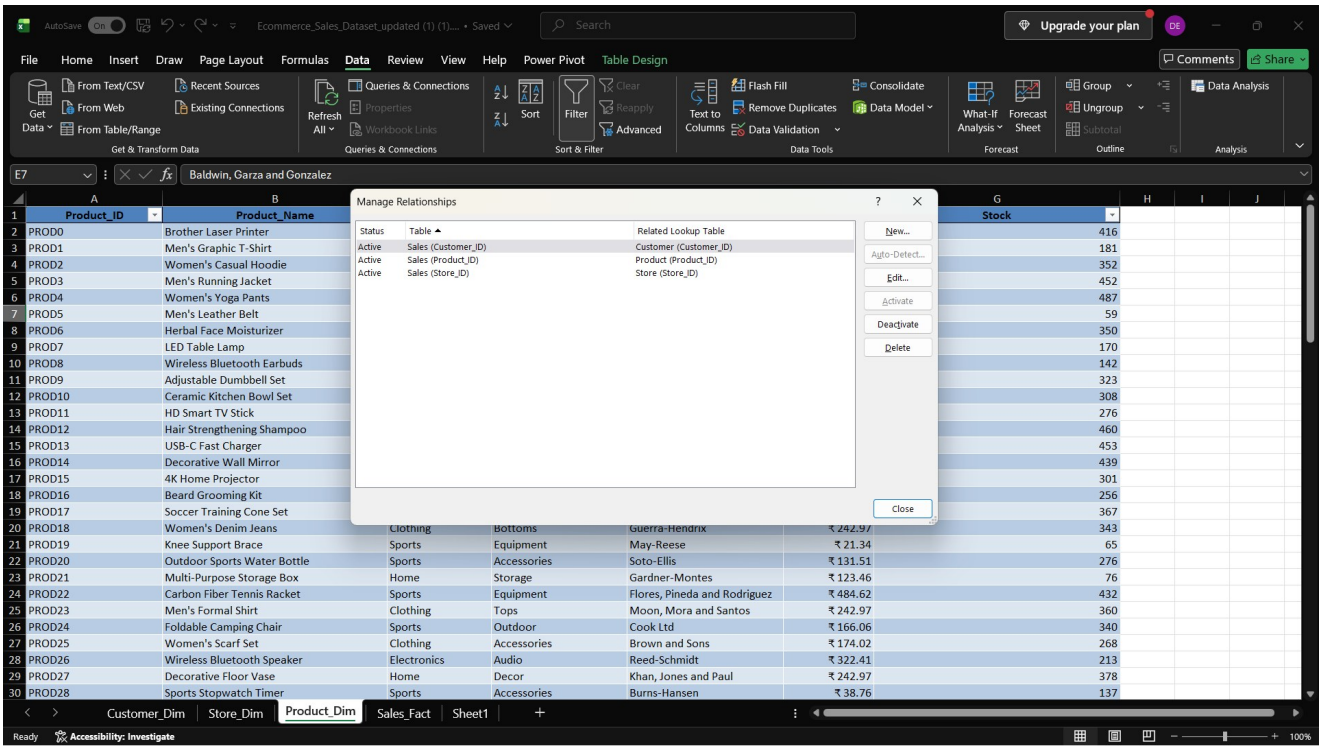


Figure 13: Relationships between Sales, Customer, Product and Store tables

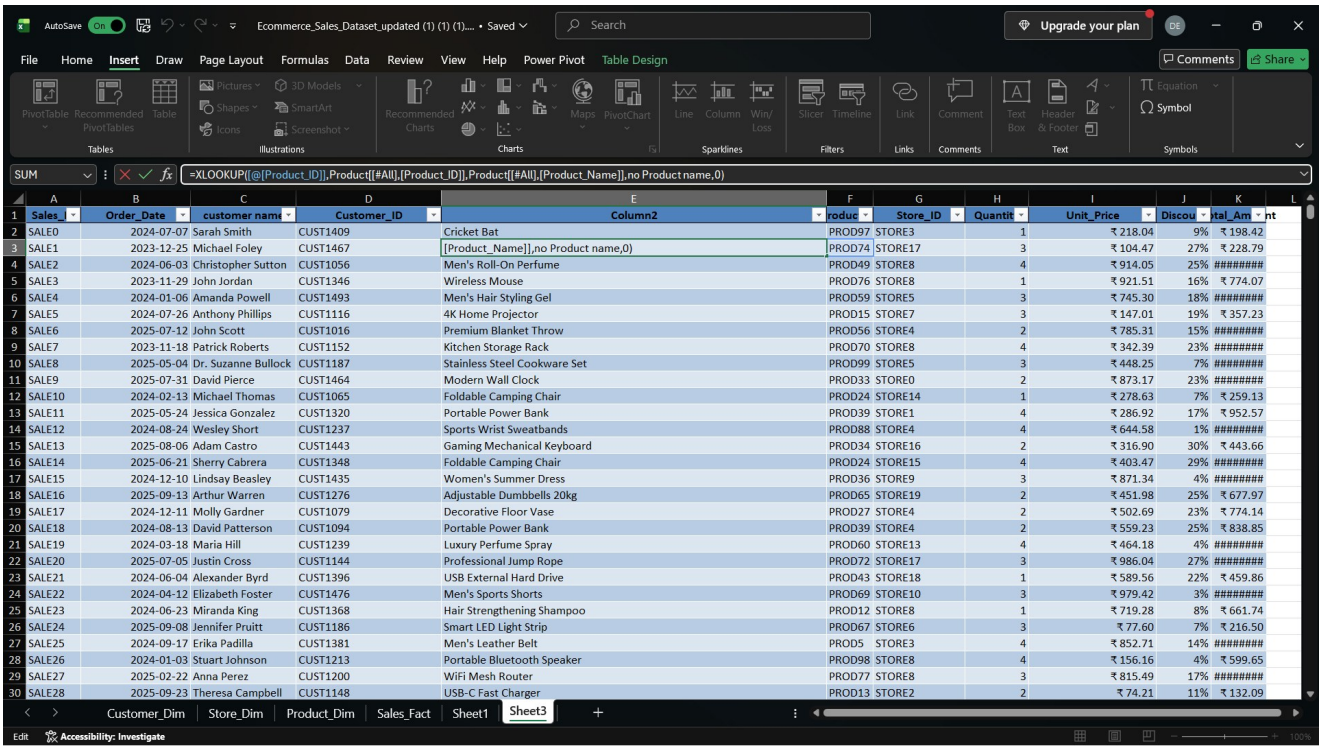


Figure 14: XLOOKUP used to bring Product_Name into the sales sheet

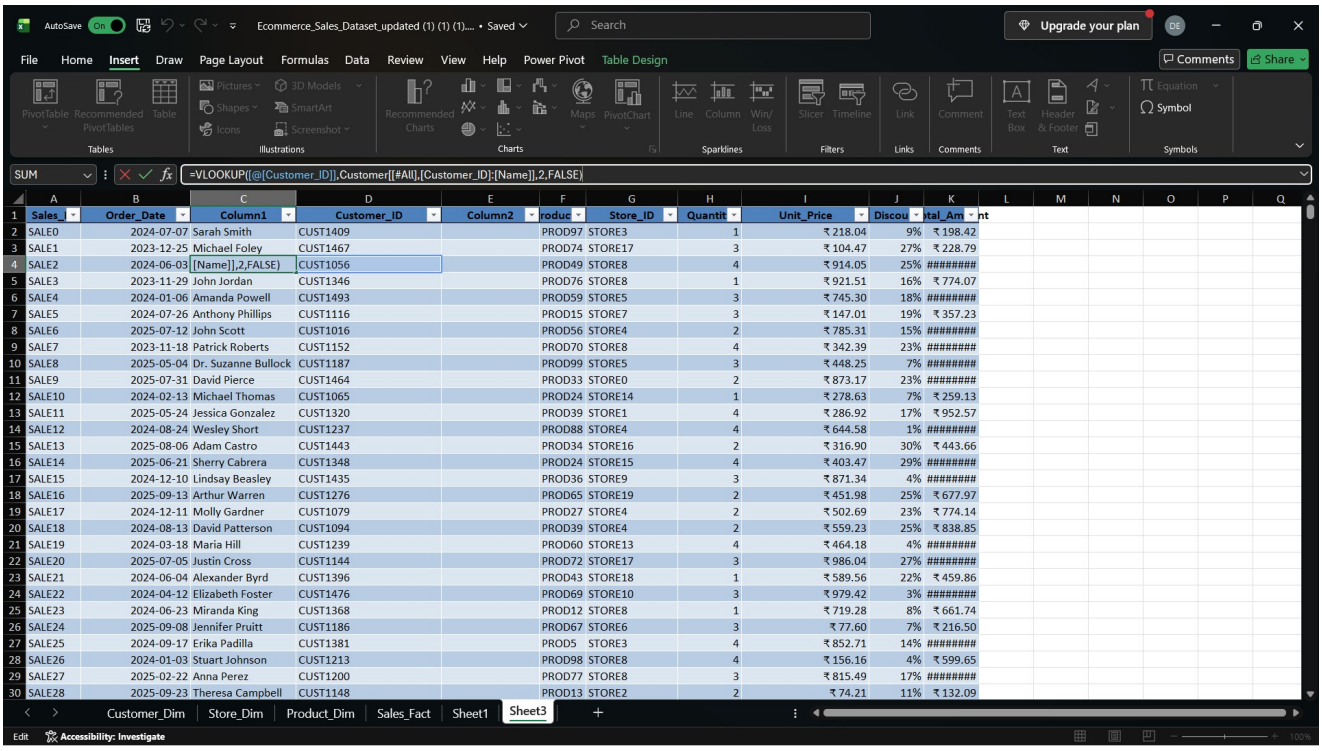


Figure 15: XLOOKUP used to bring customer Name into the sales sheet

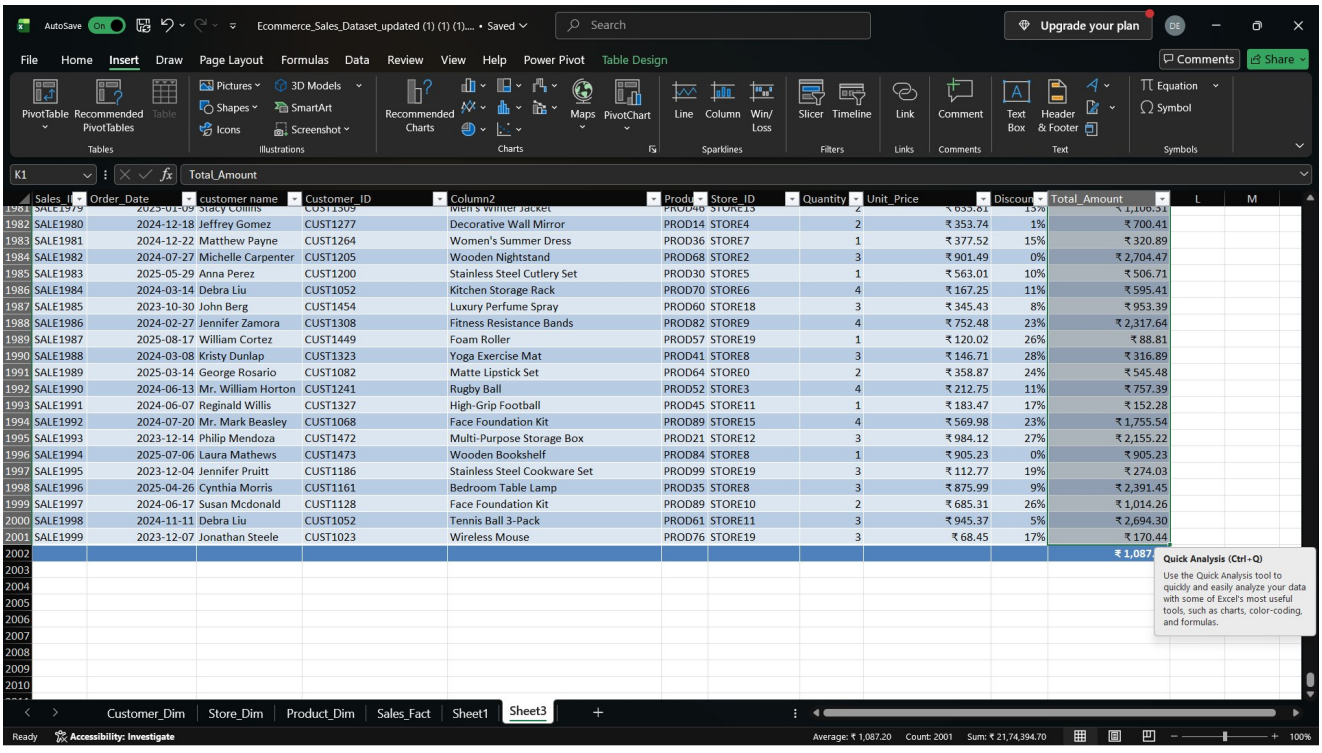


Figure 16: Total_Amount column selected for the total row

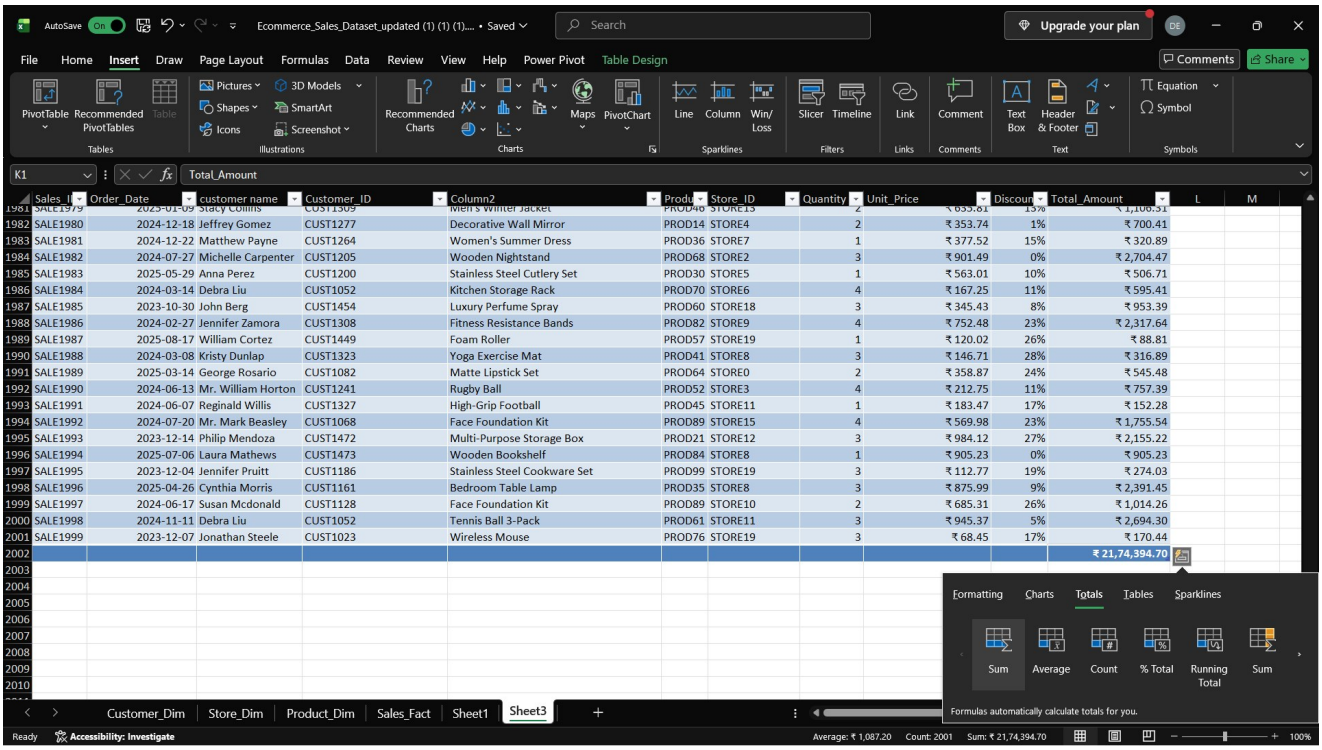


Figure 17: Total added using Quick Analysis (total sales is Rs. 21,74,394.70)

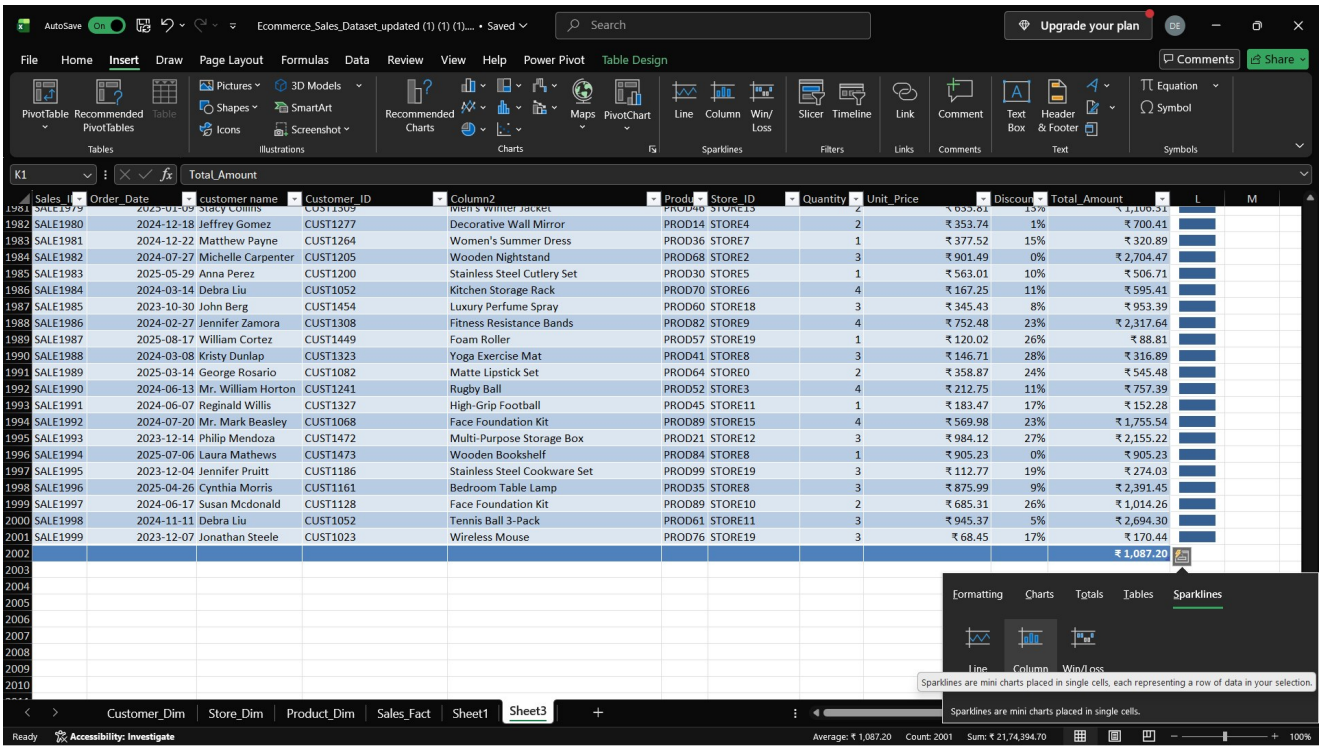


Figure 18: Sparklines added next to the Total_Amount column

Step 3: Descriptive Analysis (what happened)

Descriptive analysis shows what has already happened in the sales data. I used Descriptive Statistics on Total_Amount, and made pivot tables for sales by category and by region. The total sales is Rs. 21.74 lakh from 2000 orders, and the average order is about Rs. 1087.

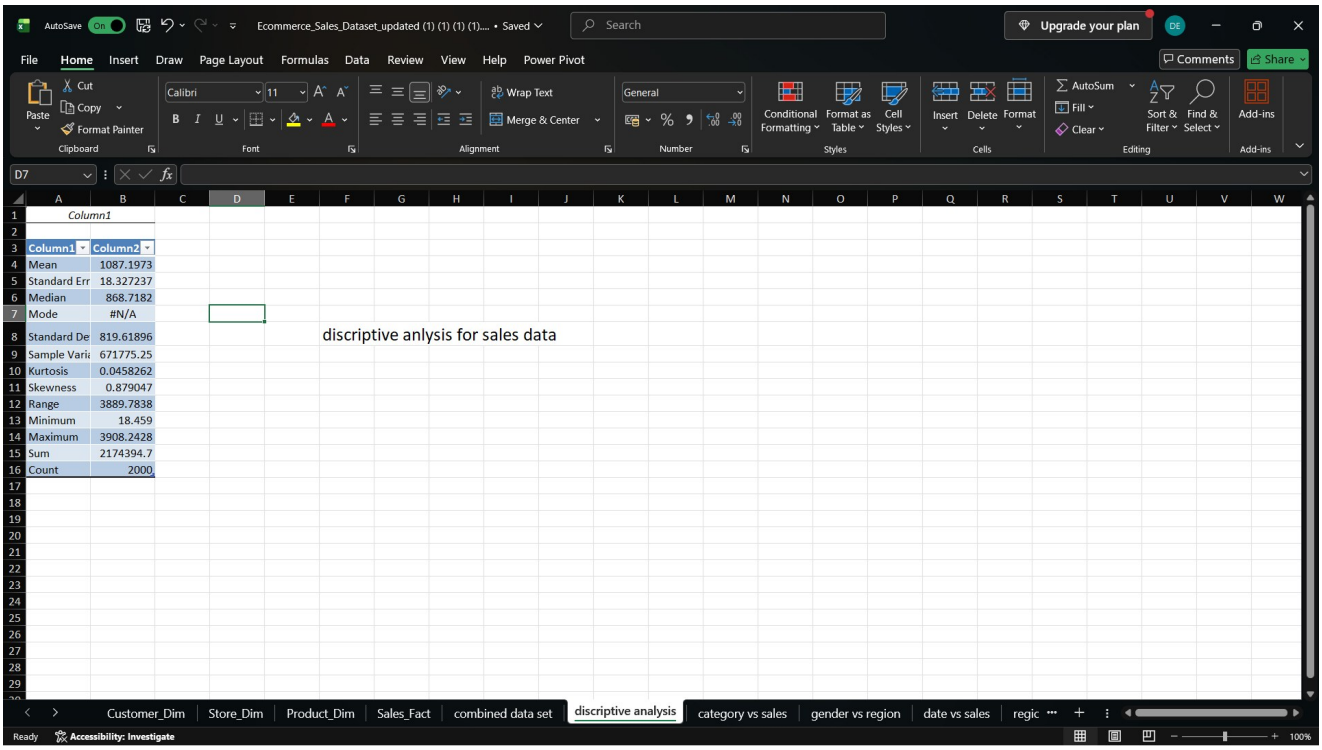


Figure 19: Descriptive statistics of Total_Amount (mean, median, standard deviation and more)

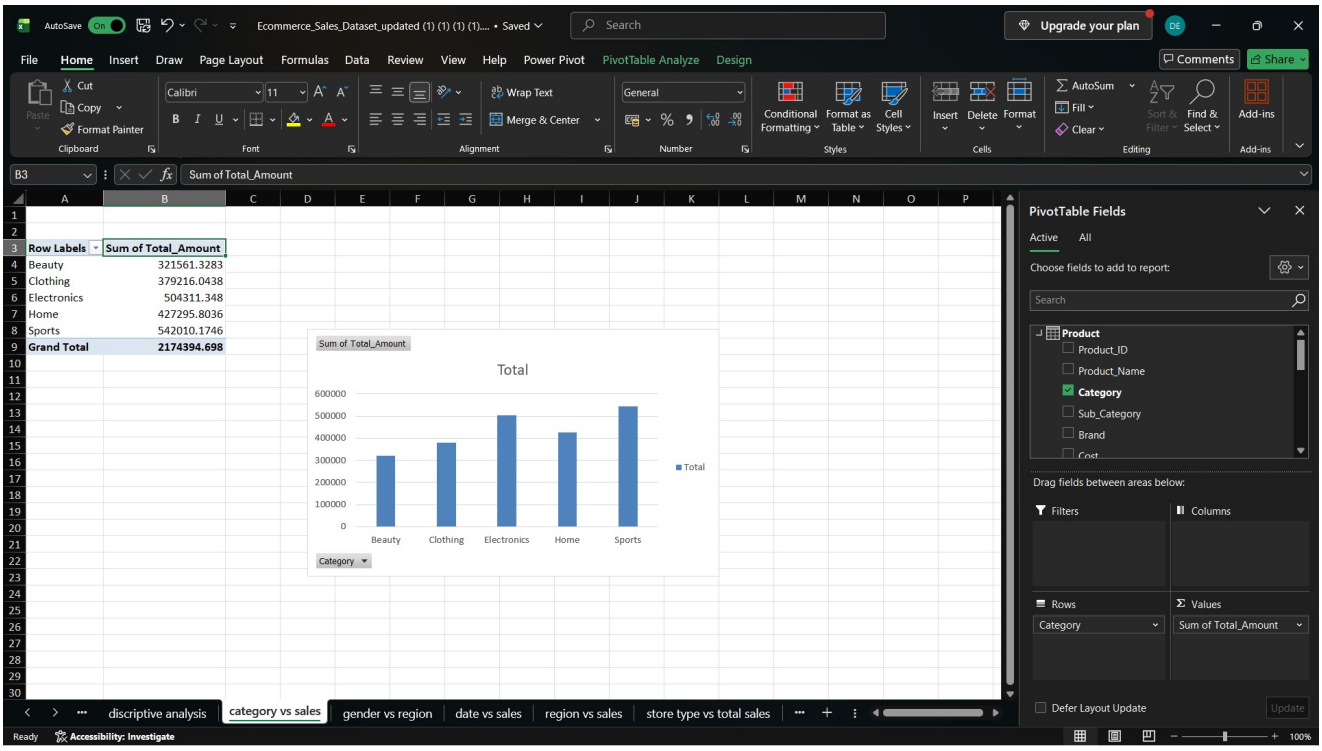


Figure 20: Pivot table and column chart of total sales by category

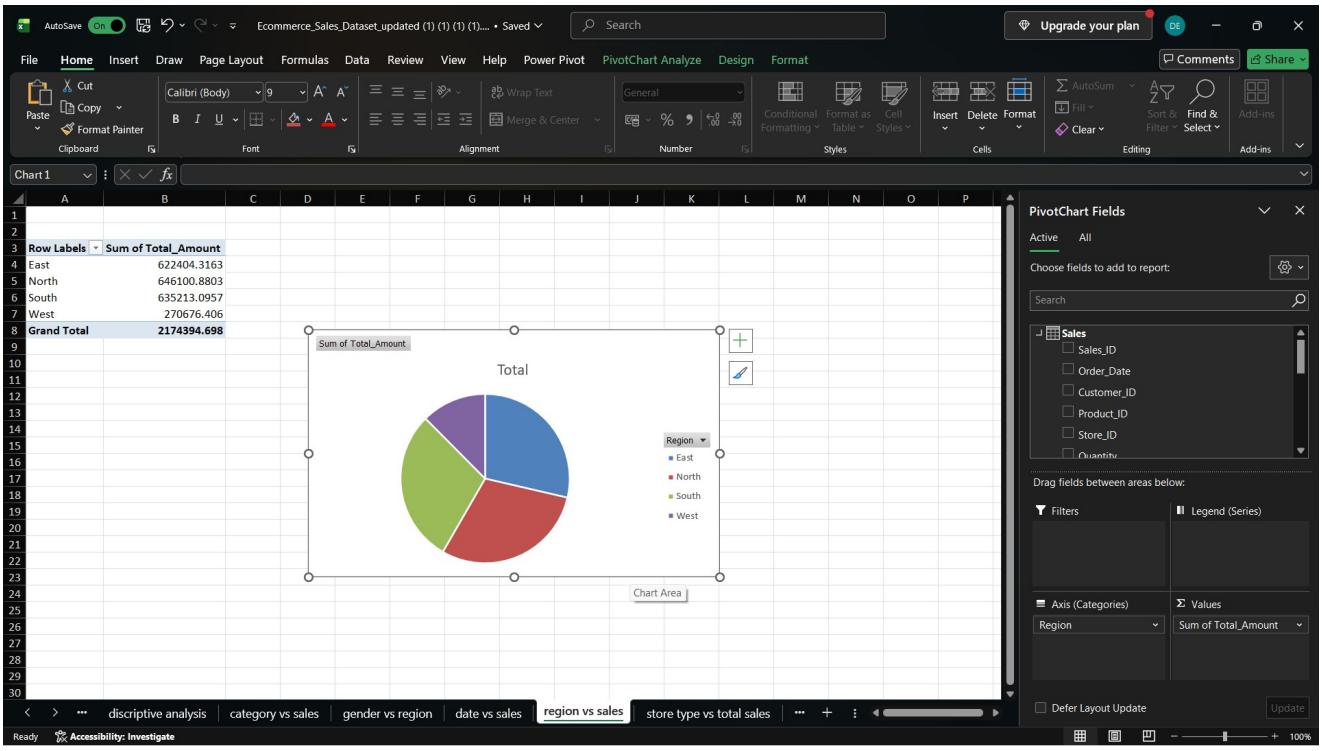


Figure 21: Pivot table and pie chart of total sales by region

Step 4: Diagnostic Analysis (why it happened)

Diagnostic analysis tries to find the reason behind the numbers. I split the sales by gender and region, and by store type and category. This shows which customer groups and which stores are pushing the sales up or down.

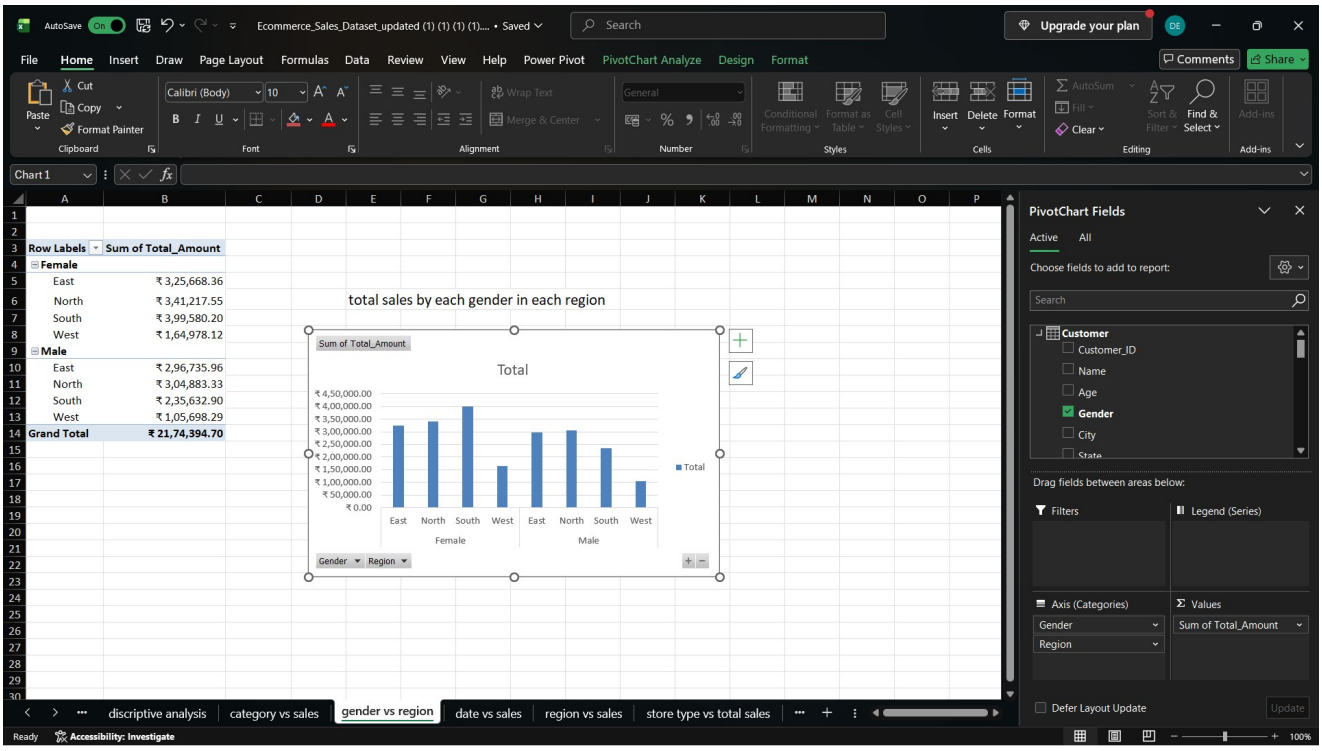


Figure 22: Total sales by each gender in each region

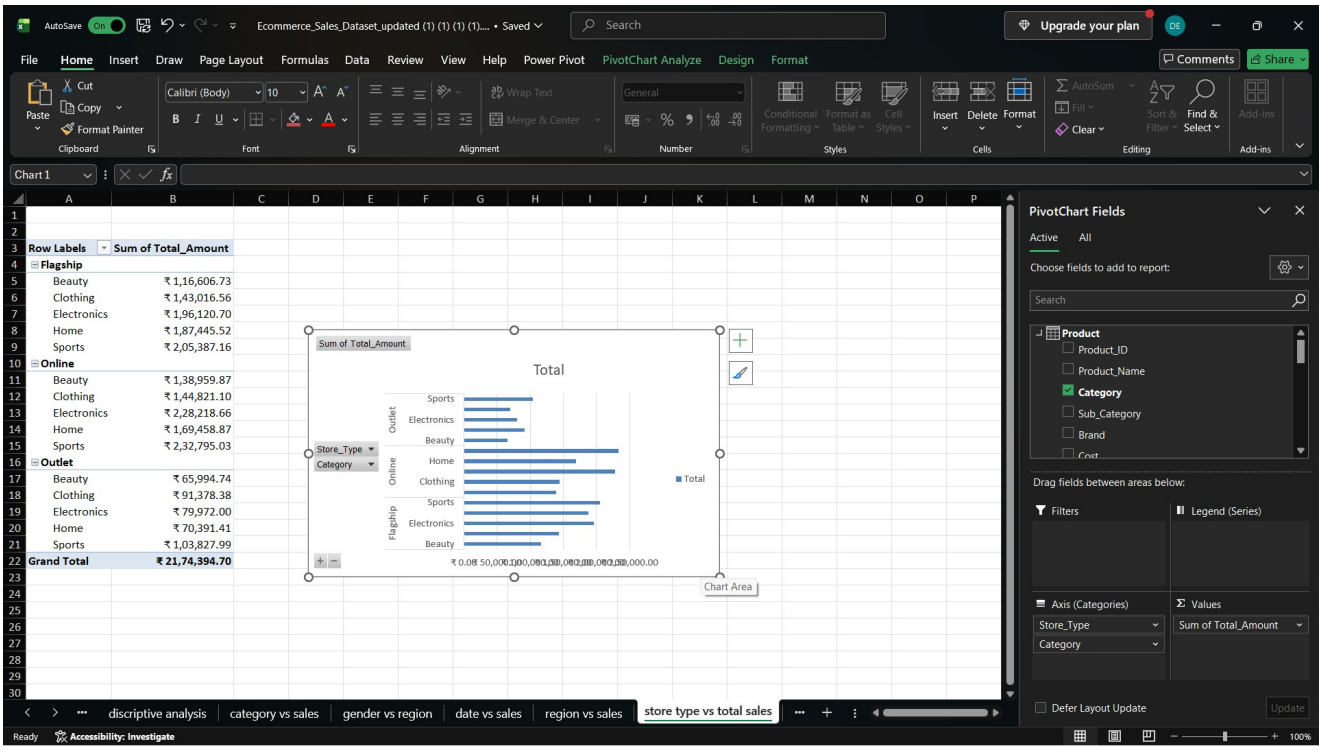


Figure 23: Total sales by store type and category

Step 5: Predictive Analysis (what may happen)

Predictive analysis uses past data to guess what may come next. I made a monthly sales trend from October 2023 to October 2025 using date grouping in the pivot table. The line goes up and down but stays mostly around 90 thousand to 1 lakh a month, so the coming months may look similar.

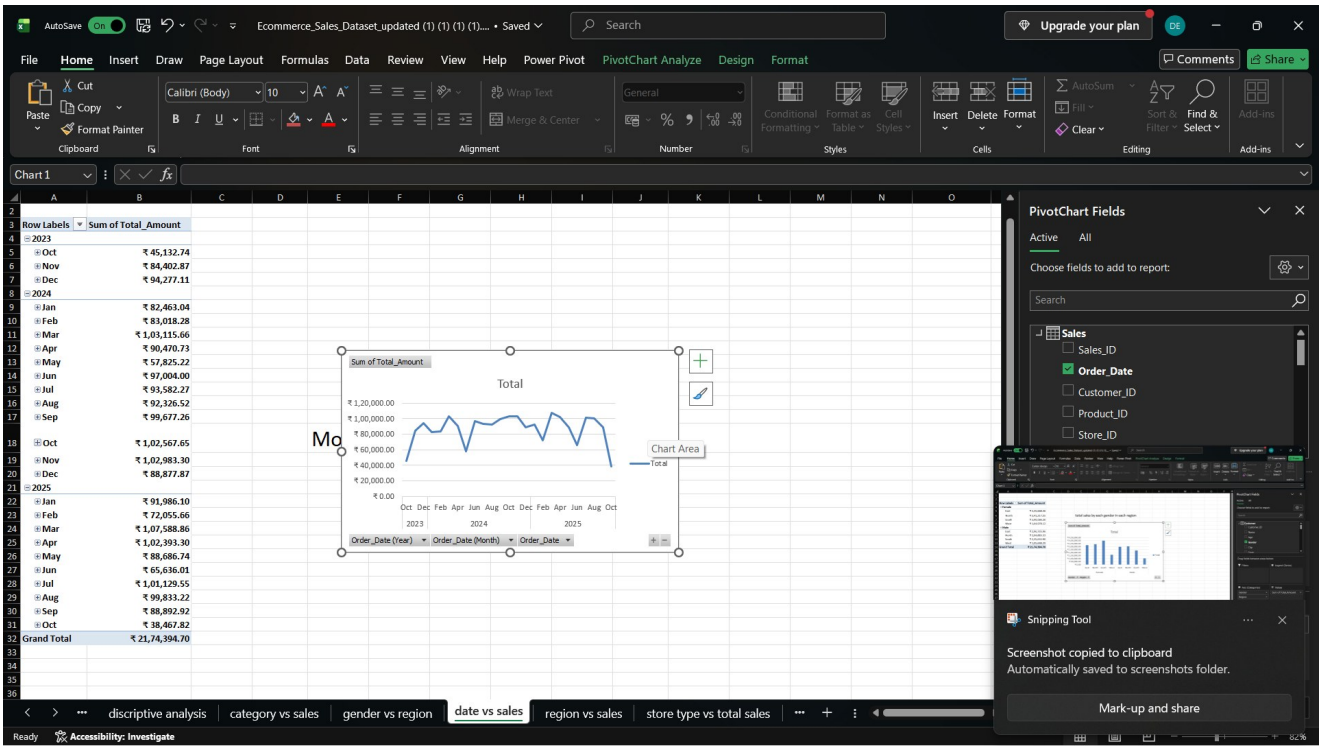


Figure 24: Monthly sales trend (pivot table and line chart)

Step 6: Prescriptive Analysis (what to do)

Prescriptive analysis tells what action we should take. I made a top 10 products pivot using the value filter to find the best sellers. Based on this, the company should keep enough stock of these products and give more offers to the weaker categories and regions.

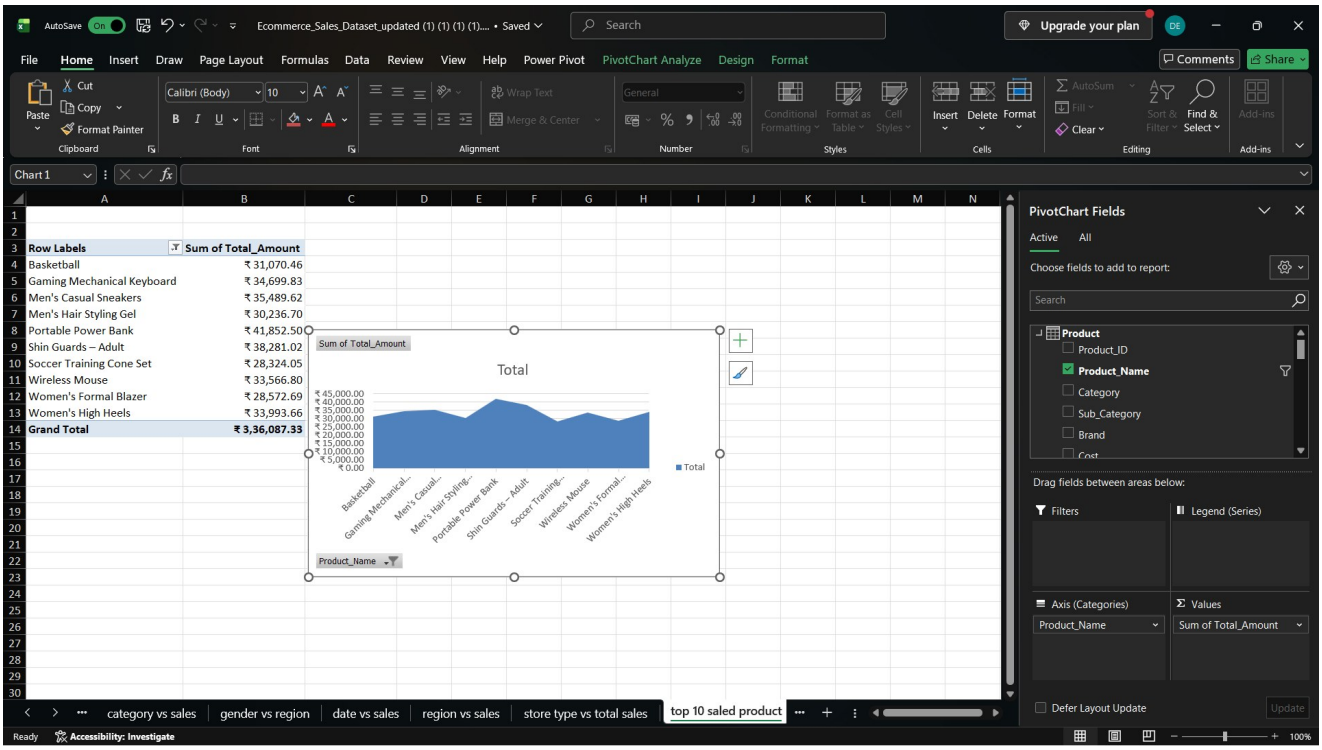


Figure 25: Top 10 best-selling products by revenue

Step 7: Dashboard

Finally I put all the charts on one sheet and added slicers for Category, Region, City, Store_Type and Gender. When a slicer is clicked, all the charts change together. This makes it easy to look at sales for one region, category or store type.

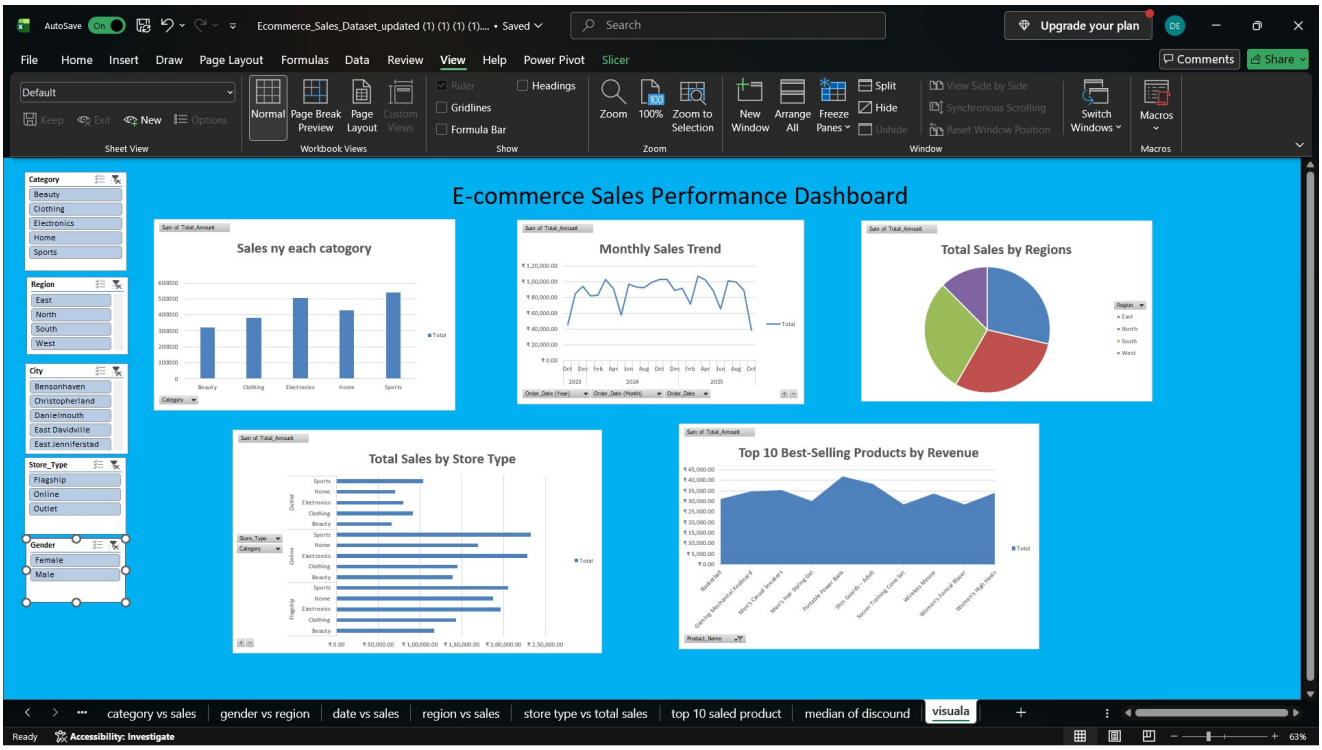


Figure 26: E-commerce Sales Performance Dashboard with slicers

Insights from the Dashboard and Pivot Tables

1. Total sales is about Rs. 21.74 lakh from 2000 orders, and the average order is about Rs. 1087. The median is Rs. 869, so a few big orders are pulling the average up.
2. Sports (about Rs. 5.42 lakh) and Electronics (about Rs. 5.04 lakh) earn the most, and Beauty earns the least (about Rs. 3.22 lakh).
3. North, South and East are almost equal at about Rs. 6.2 to 6.5 lakh each, but the West is very low at about Rs. 2.71 lakh, which is only around 12% of total sales.
4. Female customers spend more than male customers in every region, and the South has the highest female sales (about Rs. 4 lakh).
5. Online stores sell the most (about Rs. 9.14 lakh), then Flagship (about Rs. 8.49 lakh), and Outlet stores sell the least (about Rs. 4.12 lakh).
6. Monthly sales stay mostly between Rs. 90 thousand and Rs. 1 lakh, with the highest month being March 2025 (about Rs. 1.08 lakh), and Portable Power Bank is the top selling product at about Rs. 41,850.